10 **(a)** Express $\dfrac{1}{(4x+1)(x+1)}$ in partial fractions. **[3]**

(b) A curve passes through the point (0, 2) and satisfies the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y}{(4x+1)(x+1)},$$

for $x > -\frac{1}{4}$.

Show by integration that $y = A\left(\dfrac{4x+1}{x+1}\right)^{B}$ where A and B are constants to be determined. **[6]**

10(a)

10(b)

(answer space continued on next page)

10(b)	**(continued)**
	$A =$
	$B =$